

## Supplementary Materials

### Drug-tolerant persists to TRAIL emerge from a dose-dependent surface in a cell-state continuum of sensitivity.

5 Giada Fiandaca, Marielle P  r  , Kelian Bonhomme, Madalena Chaves, J  r  mie Roux

### Supplementary figures

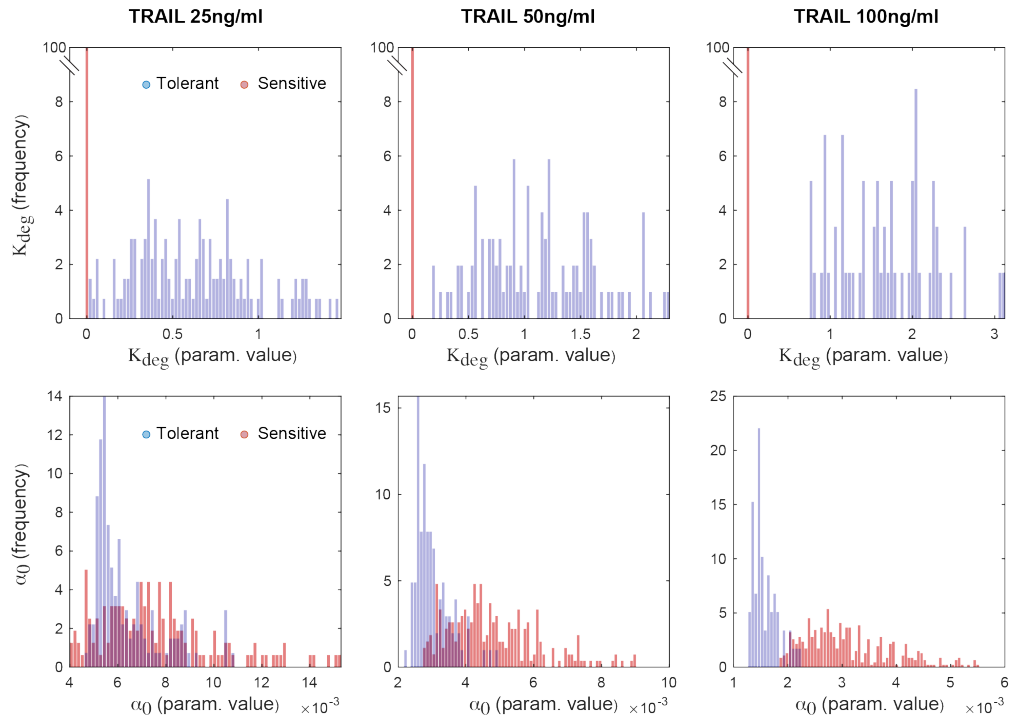


Figure S1: Parameter distributions for the core model developed in [1] for three different TRAIL doses, 25, 50, and 100 ng/ml (left to right). **(A)** Degradation rate constant  $K_{deg}$  and **(B)** Caspase 8 activation rate constant  $\alpha_0$ . Distributions differ markedly between sensitive (red) and tolerant (blue) cells, indicating that the core model cannot yet represent a common pathway for caspase activation in both phenotypes.

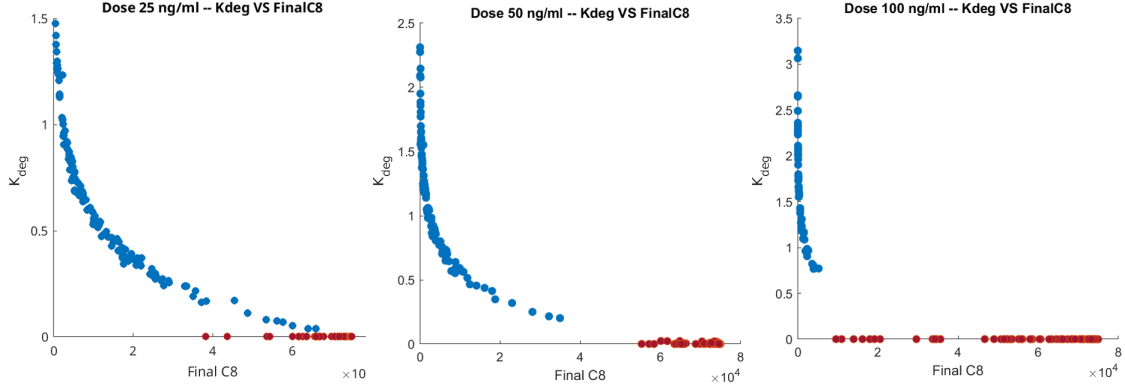


Figure S2: Relationship between parameter  $K_{deg}$  and final value of caspase 8 concentration, for the core model [1]. This relationship, almost constant for sensitive cells and a clear dependence on caspase 8 for tolerant cells, led us to consider a saturated degradation rate  $K_{deg}C8/(C8 + \alpha_{C8})$ , which considerably improved the model.

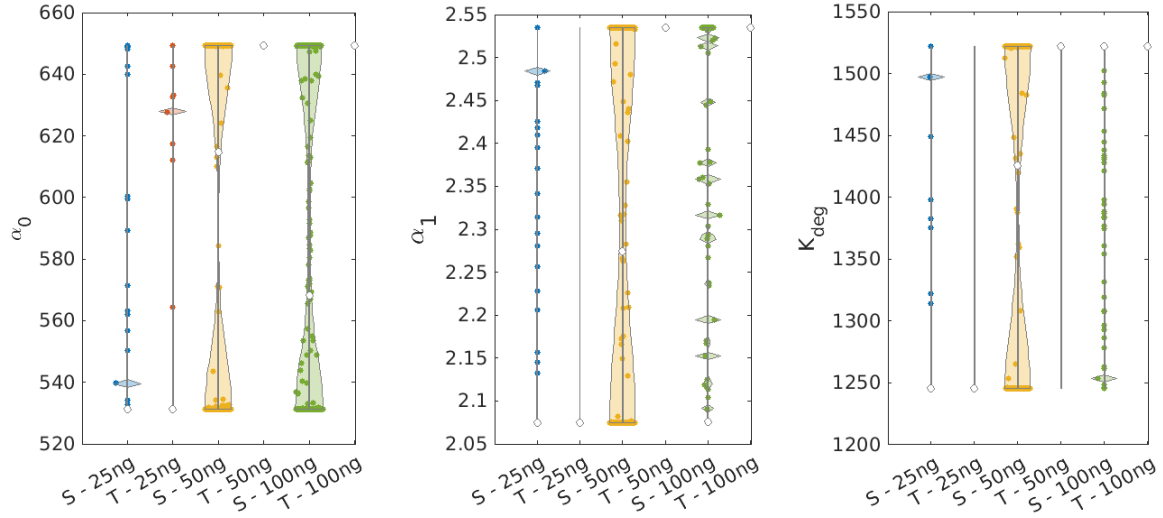
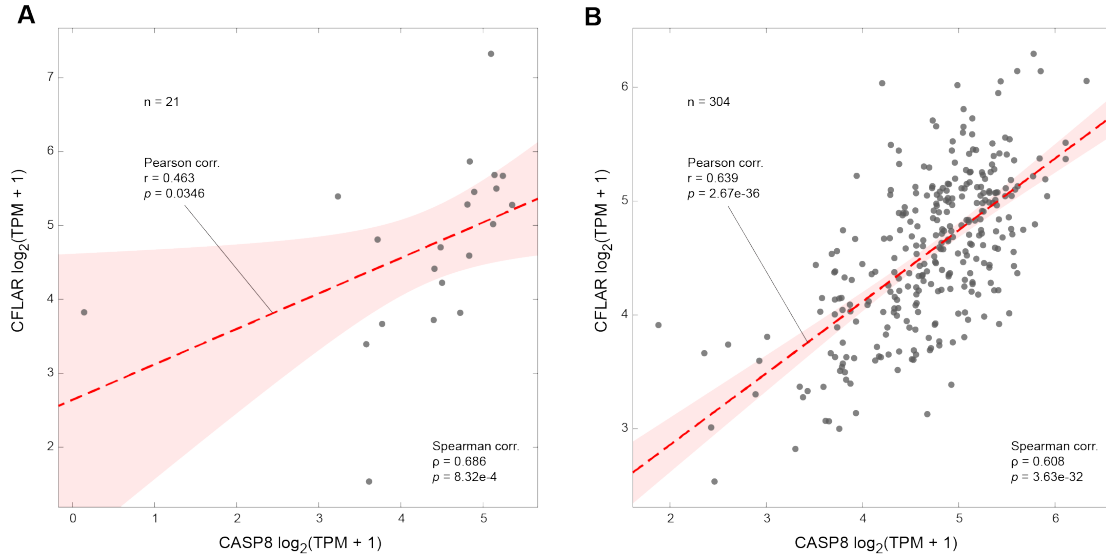


Figure S3: Violin plots showing the parameter distributions for  $\alpha_0$ ,  $\alpha_1$ ,  $K_{deg}$  (left to right, respectively), for the new model (2). Each colored dot represents the parameter for one fitted cell and each column represents a phenotype (S for sensitive or T for tolerant) and a drug dose (25, 50, or 100 ng/ml), as indicated in the horizontal axis. The dependence of the parameters on the sensitive or tolerant phenotype is now reduced relative to the core model (1), but these three parameters characterize the key terms for caspase 8 regulation and are still likely to contain some information on the distinguishing features between phenotypes and drug doses. This is apparent as the distributions are more concentrated in a small interval for tolerant phenotype.



**Figure S4: Relationship between CASP8 and CFLAR mRNA expression in cervical cancer models and tumors.** (A) Scatter plot showing CASP8 and CFLAR expression levels (transcripts per million, TPM) across cervical cancer cell lines from the DepMap dataset (Cervix lineage). Each point represents one cell line. The red dashed line indicates the linear regression fit, and the shaded area represents the 95% confidence interval. (B) Scatter plot showing CASP8 and CFLAR expression levels (transcripts per millions, TPM) in primary cervical squamous cell carcinoma (CESC) tumor samples from OncoDB (TCGA). Each point represents one patient sample. The red dashed line indicates the linear regression fit, and the shaded area represents the 95% confidence interval. Together, these analyses show that CASP8 and CFLAR expression levels are significantly correlated in both cell line and tumor datasets, providing a population-level observation consistent with, but not sufficient to validate, the coupling suggested by the model in our single-cell TRAIL response system. Expression values are presented on a  $\log_2(TPM + 1)$  scale, Pearson and Spearman correlation coefficients are shown in each panel.

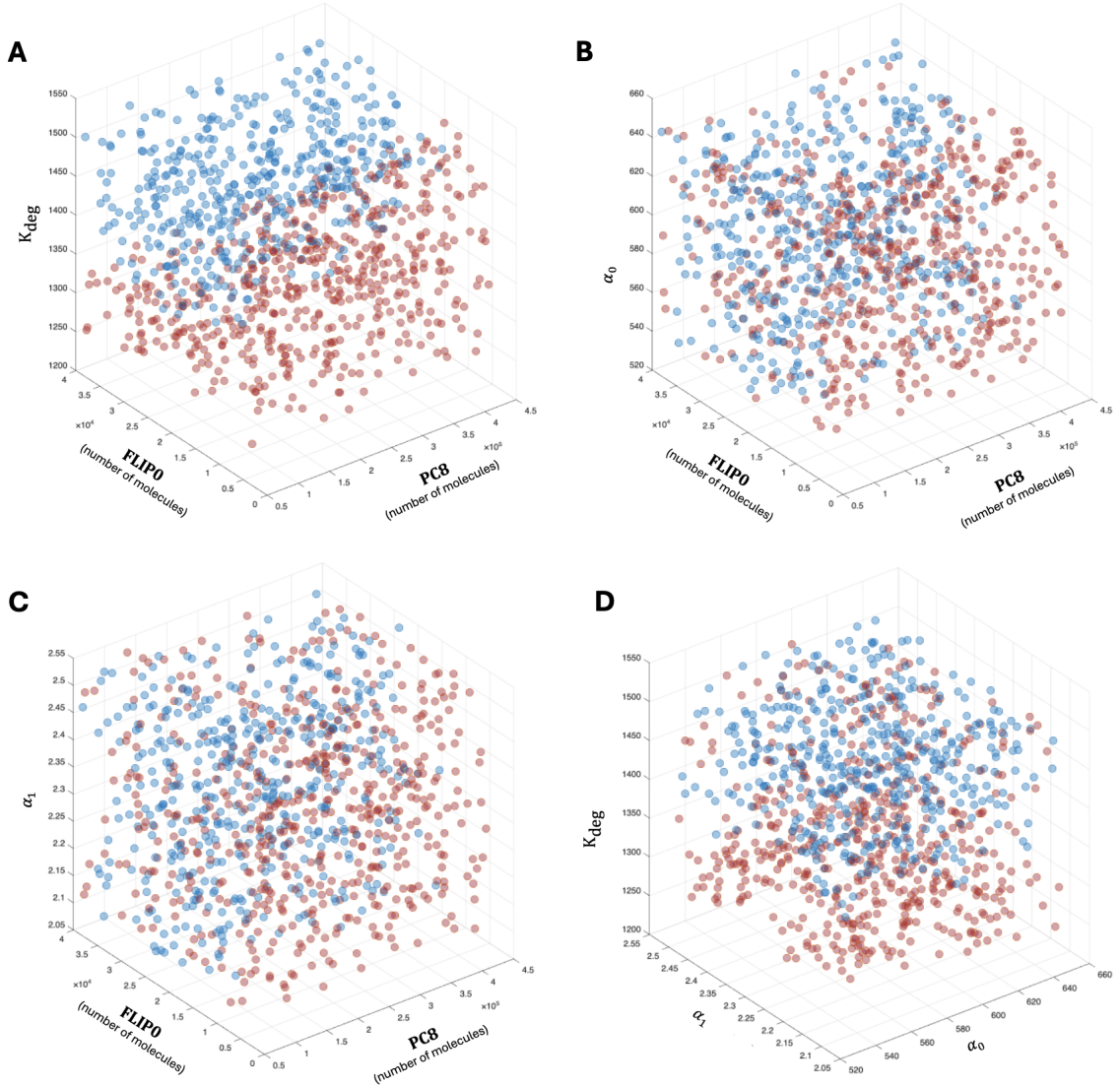


Figure S5: State-space sampled cell states represented in the  $(pC8(0), FLIP(0), K_{deg})$  (panel A),  $(pC8(0), FLIP(0), \alpha_0)$  (panel B),  $(pC8(0), FLIP(0), \alpha_1)$  (panel C) and  $(\alpha_0, \alpha_1, K_{deg})$  (panel D) state space and colored according to their predicted fate (red: sensitive, blue: tolerant) using the fixed threshold  $\theta^{ref}$ . Comparing the panels, we concluded that the triplet  $(pC8(0), FLIP(0), K_{deg})$  provides the most informative set of predictors for determining cell fate after drug administration. In particular, the pivotal role of  $K_{deg}$  emerges clearly from panels A and D. Conversely,  $\alpha_0$  and  $\alpha_1$  appear broadly distributed across both phenotypes, indicating that the two subpopulations are largely indistinguishable with respect to the caspase-8 activation parameters.

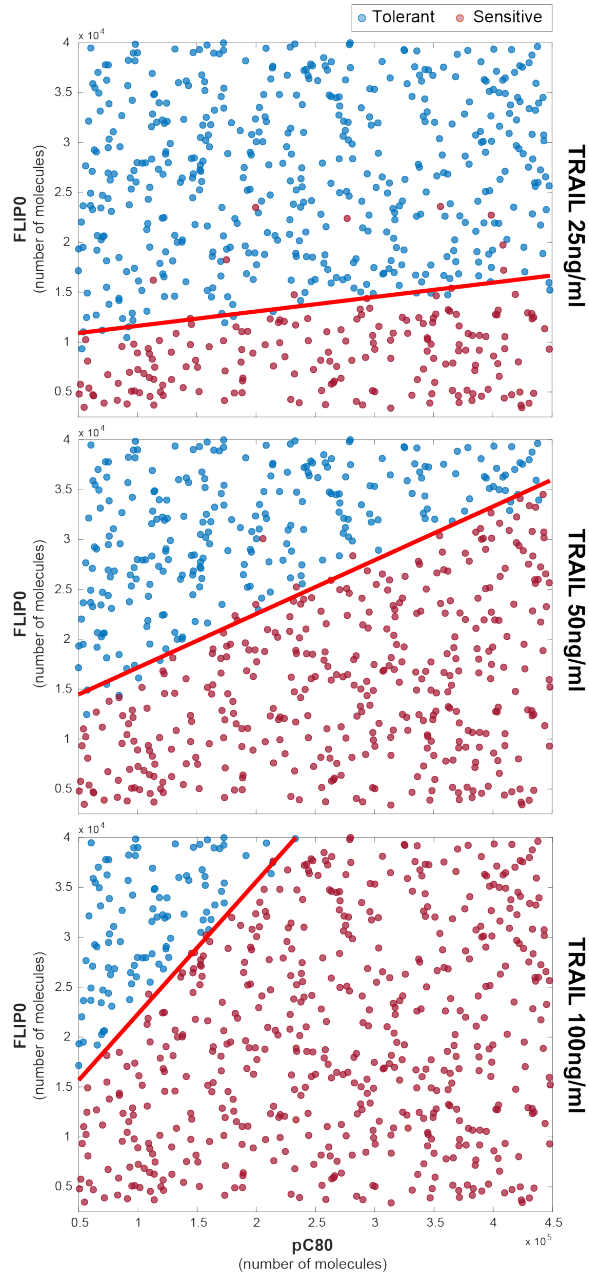


Figure S6: Dose-dependent cell-fate hyperplane defines a drug sensitivity threshold in a parameter landscape of cell death regulators. Projection of the dose-dependent hyperplane into the two-dimensional space ( $pC8(0)$ ,  $FLIP(0)$ ), for drug doses 25, 50, and 100 ng/ml, respectively. Red and blue points correspond to state-space sampled cell states classified as sensitive or tolerant using  $\theta^{ref}$ . The red line in each panel represents the projection of the fitted hyperplane, which separates the two classes. As dose increases, the line shifts and rotates, reflecting the increasing FLIP levels that can still be tolerated before crossing into sensitivity. Note that these red lines appropriately compare to the analytical estimates, as illustrated in Fig. 6 of the paper, thus further validating our model.

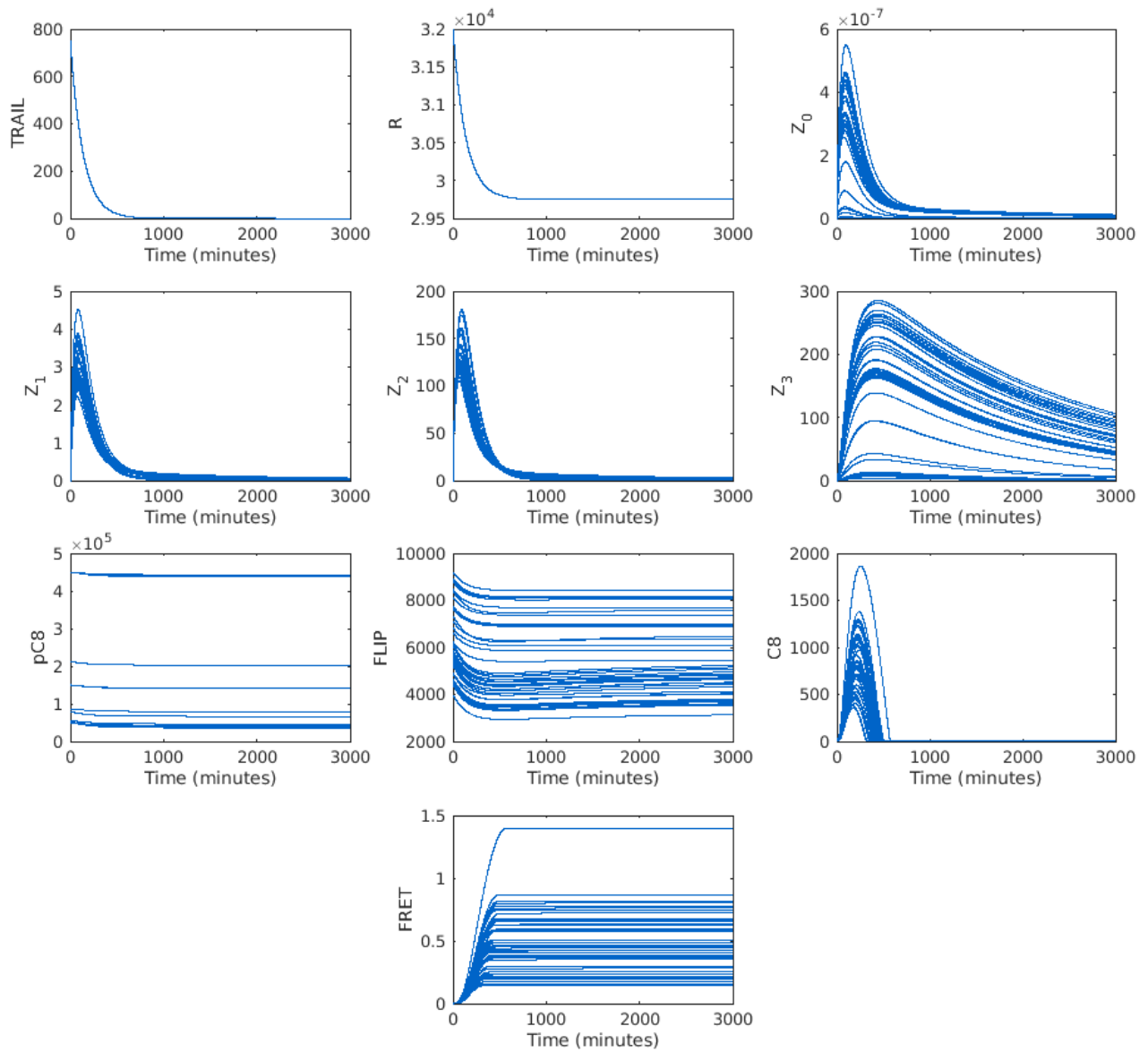


Figure S7: Long-time behavior of the extended model variables. Simulations of model (2) showing the time evolution of all variables  $T$ ,  $R$ ,  $Z_0$ ,  $Z_1$ ,  $Z_2$ ,  $Z_3$ ,  $pC8$ ,  $FLIP$ ,  $C8$ , and  $FRET$ . The curves correspond to all solutions of the fitted cells for the TRAIL dose of 50 ng/ml.

## Steady state analysis of the extended apoptosis model

**Equilibria of the extended model** The equilibria (or steady states) of system (2) can be computed as the solution  $\bar{X} = (\bar{X}_1, \bar{X}_2, \dots, \bar{X}_{10})$  of the equations  $\dot{X}_i = f_i(\bar{X}) = 0$  for all variables  $X_i$ . After some algebraic computations, the equilibria of the system can be characterized by:

$$\begin{aligned} & \bar{Z}_1 = 0, \quad \bar{Z}_2 = 0, \quad \bar{C}8 = 0, \\ \text{and:} \quad & \bar{Z}_1 = 0, \quad \text{implies} \quad \bar{Z}_0 = 0 \text{ or } p\bar{C}8 = 0 \end{aligned}$$

Since  $pC8(0)$  is present in very large amounts (much larger than  $T(0)$ ,  $R(0)$ , or  $FLIP(0)$ ), we expect the equilibria of the system will satisfy  $\bar{Z}_0 = 0$ , which in turn leads to:

$$\bar{Z}_0 = 0, \quad \text{implies} \quad \bar{Z}_3 = 0 \quad \text{and} \quad (\bar{T} = 0 \text{ or } \bar{R} = 0).$$

- 5 Since  $R(0)$  is much larger than  $T(0)$ , we expect  $\bar{T} = 0$ . The remaining quantities cannot be uniquely determined from the equations  $f_i(\bar{X}) = 0$ . But they will be positive and depend on the initial conditions:

$$\bar{R} > 0, \quad p\bar{C}8 > 0, \quad \bar{F}LIP > 0, \quad \bar{F}RET > 0. \quad (1)$$

**Interpretation of asymptotic behavior** This analysis is illustrated in Figure S7, which shows the behavior of the model at long simulation times, as each variable approaches its equilibrium value.

- 10 As expected, all variables but the four listed in equation (1) approach 0. This behavior can be interpreted as a follow up of TRAIL depletion: as its molecules are used up or sequestered, formation of all complexes  $Z_i$  naturally stops and any remaining amounts are degraded or used up to activate  $C8$ . When no more complexes are available, caspase-8 can only degrade or activate  $FRET$ . As  $C8$  tends to zero, eventually  $FRET$  also stops increasing and reaches a constant level.
- 15 The solutions shown in Figure S7 represent all cells fitted for the TRAIL dose of 50 ng/ml. It is clear that the four variables in (1) reach different values depending on the initial conditions  $pC8(0)$  and  $FLIP(0)$  (we recall that  $T(0)$  and  $R(0)$  are equal for all cells, which explains why  $\bar{R}$  is likewise equal for all cells).

- 20 **Equilibria vs. phenotypes** As a final remark, note that our extended apoptosis model admits a single type of equilibria, with depleted TRAIL, complexes, and caspase-8, a state that represents a "cell" long after TRAIL treatment is exhausted. If the cell survived, it has therefore reverted to a *similar state as before the treatment*, with some large amounts of  $R$ ,  $pC8$  and  $FLIP$  and other elements at zero. In other words, after several hours, the cell is ready to respond to new signaling,
- 25 another realistic feature of our model.

We also note that, in any case, these equilibria are not to be viewed as sensitive or tolerant phenotypes. In this model, by construction, these two phenotypes are defined by the transient behavior of the model, as characterized by the successive steps in Box 1.

## References

- 30 [1] Marielle Péré, Madalena Chaves, and Jérémie Roux. "Core models of receptor reactions evaluate basic pathway designs enabling heterogeneous commitments to apoptosis". In: *18th Conf. Computational Methods in Systems Biology (CMSB 2020)*. Konstanz, Germany, Sept. 2020.